Armadale Kalamunda Group

Group B Streptococcus Guideline

1. Purpose

- To ensure women who are Group B Streptococcus (GBS) carriers are identified and treated appropriately.
- To provide guidance for Obstetricians, Paediatricians, General Practitioners and Midwives on the prevention of early onset neonatal Group B Streptococcus (OEGBS).

2. Background

GBS is a common bacterium which may be found asymptomatically in the GI tract, vagina and urethra. GBS has emerged as an important and frequent cause of perinatal morbidity and mortality. Colonisation of the lower genital tract is common with 20% - 24% of pregnant women having positive vaginal or rectal cultures. GBS colonisation of the infant is acquired during labour and delivery from the maternal genital tract in approximately 40% - 70% of culture positive women. In the absence of antibiotic prophylaxis, approximately 1% - 2% of infants born to GBS colonised mothers will develop early onset sepsis characterised by pneumonia and meningitis. Screening and intrapartum antibiotics do not prevent late onset GBS infection (from 1 week to 3 months of age).

3. Risk Factors for Neonatal EOGBS

- Clinical diagnosis of chorioamnionitis
- other twin with current EOGBS
- Antenatal GBS colonisation of the vagina or rectum
- Antenatal GBS bacteriuria
- Previous infant with early or late inset GBS disease
- Maternal temperature ≥38 degrees during the intrapartum period
- An unknown culture result, or no screening in pregnancy and any of the following are present:
  - the onset of premature labour < 37 weeks gestation[1]
  - rupture membranes for ≥ 18 hours
  - an intrapartum fever of ≥ 38C.

4. Process

Early in pregnancy all women should be offered appropriate information concerning GBS, screening for the disease and prevention strategies if found to be present.

Offer GBS screening via a combined low vaginal-rectal swabs at:

- At 35–37 weeks gestation

- or 3-5 weeks prior to anticipated birth in high-risk pregnancy such as multiple pregnancy

Women undergoing a planned caesarean section (in the absence of labour and ruptured membranes) do not require prophylaxis for GBS regardless of GBS colonisation status. The risk of neonatal GBS is extremely low in this circumstance.

4.1 Antepartum Treatment

- Is only required if the woman has a GBS positive bacteriuria.
- There is no need to give antibiotic treatment of positive GBS colonisation as it is ineffective in eradicating carriage.

4.2 Intrapartum Prophylaxis

The use of intrapartum prophylaxis with antibiotics given to women at risk of transmission of GBS to their newborns has been shown to reduce the incidence of early onset sepsis by 86-89%.

Is required for

- A positive GBS culture
- GBS positive with ruptured membranes before caesarean section.
- An unknown culture at the time of labour where labour is preterm (< 37 weeks), or membranes ruptured for ≥18 hours, or intrapartum fever of ≥38 degrees Celsius is present.
- A history of an infant with GBS disease regardless of culture
- Symptomatic or asymptomatic GBS bacteriuria during pregnancy. Urinary tract infections caused by GBS complicate 2%-4% of pregnancies

Intrapartum **antibiotic prophylaxis is not required** when Caesarean birth is performed before the onset of labour on a woman with intact amniotic membranes, regardless of GBS colonisation status or Gestational age.

NB: routine Caesarean section antibiotic prophylaxis is still required independent of GBS status or gestational age.

The recommended intrapartum prophylaxis for GBS is IV Benzylpenicillin 3.0gms initially (as a loading dose), then 1.8gms IV every 4 hours until delivery.

Dilute in 100mls normal saline, administer over 30 minutes

If the woman is allergic to Penicillin, with a history of anaphylaxis/ angio-oedema / respiratory distress / urticarial, then:

- Clindamycin 900mg every 8 hours until delivery

Dilute in 100mls normal saline and infuse over 30 minutes

If the woman is at high risk of anaphylaxis and the isolate is known to be resistant to Clindamycin, then:

- Vancomycin 20mg/kg body weight up to 1.5g 12 hourly until delivery

Reconstitute in 10mls WFI, then dilute to at least 5mg/ml (500 mg/100mls). Infuse 500 mgs over 60 mins

Where the woman is allergic to Penicillin with no history of anaphylaxis / angio-oedema / respiratory distress or urticarial, the woman should have

- Loading dose of IV cephazolin 2gms, followed by
- Maintenance dose of IV cephazolin 2gm every 8 hours until delivery

Mix in 10mls normal saline, and administer over 3-5 mins

If the woman is at high risk of anaphylaxis and the susceptibilities of the isolates are unknown, either Clindamycin or Vancomycin can be used

Where the woman is allergic to Penicillin and Clindamycin, Vancomycin should be used. Consult with microbiologist if Vancomycin unsuitable

Screening results are applicable for five weeks.

4.3 Neonatal Symptoms of Sepsis

The presentation of sepsis in the neonate can be subtle and should be specifically considered in the following circumstances.

- Respiratory symptoms (signs of tachypnoea, grunting or oxygen requirement)
- Apnoeic episodes
- Lethargy / poor feeding
- Poor peripheral perfusion

Fever is a particularly poor marker of infection in the neonate. Variations from normal generally reflect the environment and a raised temperature may be a manifestation of dehydration.

If there has been inadequate antibiotic cover in labour of less than 4 hours prior to birth in a woman with known GBS positive status then,

- 4 hourly observations for 24 hours and CRP, +/- FBC at 24 hours old is indicated. Discharge if CRP negative

4.4 Use of Antibiotics in Neonates with Suspected GBS

- If sepsis is suspected, blood cultures and a CRP should be collected prior to administration of antibiotics.
- Penicillin and Gentamicin are the antibiotics of choice as they provide good cover against the two most common pathogens, GBS and Escherichia coli.
- Benzylpenicillin: 50mg/kg/dose 12 hourly
- Gentamicin: > 35 weeks – 4.5mg/kg/dose once every 24 hours
- If blood cultures are negative and 2 CRP results 24 hours apart are negative, antibiotics may be discontinued. Otherwise treat for 5 days or longer depending on the clinical circumstances.
- Antibiotics should be given parenterally initially (IV or IM). Depending on clinical state and CRP results at 24 hours of age, the route may be changed to oral.

5. Legislative context

The following documents are required to give effect to this policy

- Health Department of Western Australia (HDWA) Recognising and Responding to Acute Deterioration Policy (MP 0086/18)

6. Supporting information

The following documents support the implementation of this policy:

- Recognition and Response to Acute Deterioration (AKG-POL-0388-14)
- Medication Management Procedure (AKG-PRO-0255-18)
- Infection Surveillance (AKG-PRO-0206-11)

7. Compliance, monitoring and evaluation

Compliance against this guideline will be evaluated with [insert specific monitoring requirements e.g. number of unplanned readmissions].

[The Safety, Quality, Education and Innovation Department can assist with determining monitoring requirements. Compliance monitoring should include the position and/or committees responsible for ensuring compliance, monitoring of KPI's, planned audits etc.]

8. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 3 – Action 3.15 Antimicrobial stewardship
- Standard 4 – Action 4.10 Medication review
- Standard 5 – Action 5.10 Screening of risk
- Standard 8 – Action 8.10 Responding to deterioration
- Standard 8 – Action 8.13

9. References

1. RANZCOG Guidelines C-Obs 19: Maternal Group B Streptococcus in Pregnancy: screening and management , July 2019.
2. Hughes RG, Brocklehurst P, Steer PJ, Heath P, Stenson BM on behalf of the Royal College of Obstetricians and Gynaecologists. Prevention of early-onset neonatal group B streptococcal disease. Green-top Guideline No. 36. BJOG 2017;124:e280–e305.
3. KEMH Guidelines: Group B streptococcal Disease, March 2021

10. Document contact

Enquiries relating to this guideline may be directed to:

Title: Head of Department
Department: Obstetrics and Gynaecology Department
Email: Sangeeta.mallabhat@health.wa.gov.au

11. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
Version 3.1	24/05/2024	01/05/2026	Intrapartum Prophylaxis

12. Authorisation

This guideline has been authorised and issued by the Director Clinical Services.

Authorised by	Dr Tania Strickland – A/Director Clinical Services
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